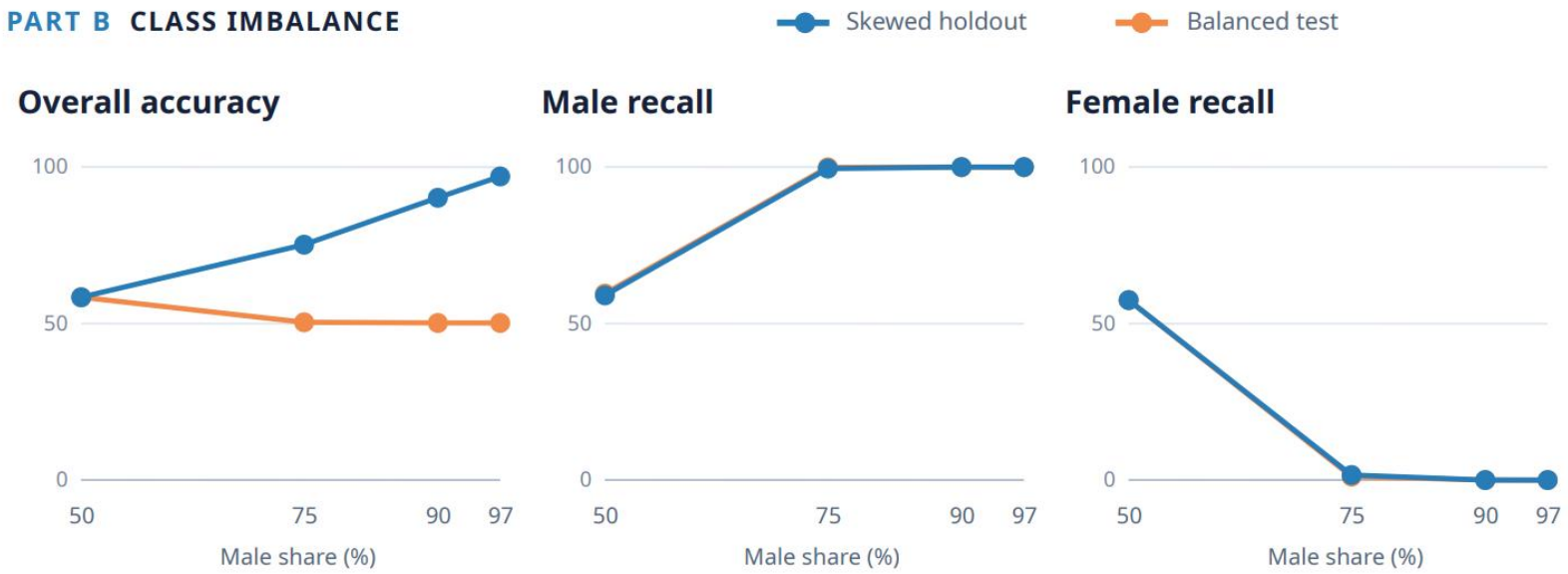
BMI 500 HW3: Model-based Bias Removal

HYPOTHESIS As the male share of training data grows, measured accuracy may rise while female recall declines.

PART B CLASS IMBALANCE



AT 97% MALE TRAINING SHARE

97%

Accuracy on skewed holdout

0%

Female recall

Balanced-test accuracy: ~50%

CONCLUSION Hypothesis partially supported: as the training set became more male-dominated, overall accuracy looked better, but female recall dropped sharply. Gaussian augmentation improved balanced-test accuracy from 50.0% to 58.46% and greatly reduced the recall gap.

Lesson learned High accuracy can hide poor performance on minority groups.